

SPENCER FREI

Research Scientist, Google DeepMind ◇ spencerfrei.github.io ◇ sfrei@google.com ◇ [Google Scholar](#)

EMPLOYMENT

- Senior Research Scientist, Google DeepMind 2025–present
- Assistant Professor, Department of Statistics, University of California, Davis 2023–2024
— Member of the Graduate Group in Computer Science
— Long-term participant at the Simons Institute Fall 2024 program on Modern Paradigms in Generalization
- Postdoctoral Fellow, Simons Institute for the Theory of Computing, UC Berkeley 2021–2023
— Mentors: Peter Bartlett and Bin Yu.
— Visitor at École polytechnique fédérale de Lausanne (EPFL), Sep. & Oct. 2022, hosted by Emmanuel Abbé.

EDUCATION

- Ph.D, Statistics, UCLA 2015–2021
— Co-advisors: Quanquan Gu and Ying Nian Wu.
- MSc., Mathematics, University of British Columbia, Vancouver 2013–2015
- BSc., Mathematics, McGill University, Montréal. 2009–2013
— First class honours.

PUBLICATIONS AND PREPRINTS

(see also [Google Scholar](#))

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1. Fangyuan Lin, **S. Frei**, Victor de la Peña. Bounded difference concentration for infinitely exchangeable sequences with applications to AI benchmark uncertainty. Preprint, arXiv:2606.17426.
 2. Alexander Tsigler, Luiz F. O. Chamon, **S. Frei**, Peter L. Bartlett. Benign overfitting and the geometry of the ridge regression solution in binary classification. Preprint, arXiv:2503.07966.
 3. Usman Anwar, Johannes von Oswald, Louis Kirsch, David Krueger, **S. Frei**. Understanding in-context learning of linear models in transformers through an adversarial lens. *TMLR 2025*.
 4. Shuning Shang, Roey Magan, Zhiwei Xu, **S. Frei**, Gal Vardi, Wei Hu. Benign overfitting in single-head attention. *NeurIPS 2025*.
 5. **S. Frei** and Gal Vardi. Trained transformer classifiers generalize and exhibit benign overfitting in-context. *ICLR 2025*.
 6. Nikhil Ghosh, **S. Frei**, Wooseok Ha, Bin Yu. The effect of SGD batch size on autoencoder learning: Sparsity, sharpness and feature learning. *JMLR 2025*.
 7. Neil Mallinar*, Austin Zane*, **S. Frei**, Bin Yu. Minimum-norm interpolation under covariate shift. *ICML 2024*.
 8. Ruiqi Zhang, **S. Frei**, Peter L. Bartlett. Trained transformers learn linear models in-context. *JMLR 2024*.
 9. Zhiwei Xu, Yutong Wang, **S. Frei**, Gal Vardi, Wei Hu. Benign overfitting and grokking in ReLU networks for XOR cluster data. *ICLR 2024*.
 10. **S. Frei***, Gal Vardi*, Peter L. Bartlett, Nati Srebro. The double-edged sword of implicit bias: Generalization vs. robustness in ReLU networks. *NeurIPS 2023*.
 11. **S. Frei***, Gal Vardi*, Peter L. Bartlett, Nathan Srebro. Benign overfitting in linear classifiers and leaky ReLU networks from KKT conditions for margin maximization. *COLT 2023*.

12. **S. Frei***, Gal Vardi*, Peter L. Bartlett, Nati Srebro, Wei Hu. Implicit bias in leaky ReLU networks trained on high-dimensional data. *ICLR 2023 (Spotlight)*.
13. **S. Frei**, Niladri Chatterji, Peter L. Bartlett. Random feature amplification: Feature learning and generalization in neural networks. *JMLR 2023*.
14. **S. Frei**, Niladri Chatterji, Peter L. Bartlett. Benign overfitting without linearity: Neural network classifiers trained by gradient descent for noisy linear data. *COLT 2022*.
15. **S. Frei***, Difan Zou*, Zixiang Chen*, Quanquan Gu. Self-training converts weak learners to strong learners in mixture models. *AISTATS 2022*.
16. **S. Frei** and Quanquan Gu. Proxy convexity: A unified framework for the analysis of neural networks trained by gradient descent. *NeurIPS 2021*.
17. Difan Zou*, **S. Frei***, Quanquan Gu. Provable robustness of adversarial training for learning halfspaces with noise. *ICML 2021*.
18. **S. Frei**, Yuan Cao, Quanquan Gu. Provable generalization of SGD-trained neural networks of any width in the presence of adversarial label noise. *ICML 2021*.
19. **S. Frei**, Yuan Cao, Quanquan Gu. Agnostic learning of halfspaces with gradient descent via soft margins. *ICML 2021 (Long Talk)*.
20. Ariana E. Anderson, Mirella Diaz-Santos, **S. Frei et al.** Hemodynamic latency is associated with reduced intelligence across the lifespan: an fMRI DCM study of aging, cerebrovascular integrity, and cognitive ability. *Brain Structure and Function*, 2020.
21. **S. Frei**, Yuan Cao, Quanquan Gu. Algorithm-dependent generalization bounds for overparameterized deep residual networks. *NeurIPS 2019*.
22. **S. Frei** and Edwin Perkins. A lower bound for p_c in range- R bond percolation in two and three dimensions. *Electronic Journal of Probability* 21(56), 2016.
23. **S. Frei**, Kathryn Lockwood, Greg Stewart, Justin Boyer, Burt S. Tilley, On thermal resistance in concentric residential geothermal heat exchangers. *Journal of Engineering Mathematics* 86(1), 2014.

* indicates equal contribution.

PRIOR INDUSTRY RESEARCH EXPERIENCE

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| Applied Scientist Intern, Amazon Alexa AI, Cambridge, MA | <i>Summer 2020</i> |
| — Worked on natural language understanding using Transformer-based multilingual language models. | |
| Student Researcher, Chatterbaby/UCLA School of Medicine, Los Angeles, CA | <i>2018–2020</i> |
| — Developed deep learning models for audio analysis of infant cries that were deployed in Chatterbaby app (100k+ downloads on Android). Employed 20 hr/week when not teaching. | |
| Biostatistical Consultant, Ritter Pharmaceuticals, Los Angeles, CA | <i>2017–2019</i> |
| — Consulted on the analysis of clinical trial data using linear regression and mixed effects models. | |
| Student Researcher, Blackthorn Therapeutics/UCLA School of Medicine, Los Angeles, CA | <i>2016–2018</i> |
| — Analyzed MRI and neuropsychiatric data using generalized linear models for a biotech company-sponsored project. Employed 20 hr/week when not teaching. | |

HONORS

- Selected for NeurIPS 2023 Tutorial, “Reconsidering Overfitting in the Age of Overparameterized Models”
- Rising Star in Machine Learning, University of Maryland, 2022.
- Best Reviewer Award: ICML 2020, ICML 2021, NeurIPS 2021, ICLR 2022.
- Dissertation Year Fellowship, UCLA, 2020–2021.
- Most Promising Computational Statistician, UCLA, 2016.
- Research Fellowship, Montréal Institut des sciences mathématiques, 2012.

TALKS

[∇] indicates virtual presentation

- “Generalization and Benign Overfitting In-Context in Trained Transformer Classifiers”
— Columbia University, Statistics Seminar *April 2025*
- “Reconsidering Overfitting in the Age of Overparameterized Models” (three hour tutorial)
— NeurIPS 2023 Tutorial, presented with Vidya Muthukumar and Fanny Yang *December 2023*
- “Learning Linear Models In-Context with Transformers”
— University of Southern California, Department of Data Sciences and Operations Seminar *October 2024*
— UCLA, Department of Statistics and Data Science Seminar *May 2024*
— École normale supérieure, INRIA SIERRA Seminar *April 2024*
— Sorbonne Université and Paris Diderot University, Statistics Seminar *April 2024*
— Apple Machine Learning Research, Cupertino *November 2023*
— University of Basel, Department of Mathematics and Computer Science Seminar *November 2023*
— University of Oxford, Computational Statistics and Machine Learning Seminar *October 2023*
— University of Cambridge, Machine Learning Group *October 2023*
— Google DeepMind, London *October 2023*
— Imperial College London, Imperial + AI Seminar *October 2023*
— The Alan Turing Institute, London, Rough Paths Interest Group[∇] *October 2023*
— Stanford University, Tengyu Ma group meeting *September 2023*
— Google Research, In-Context Learning Reading Group[∇] *August 2023*
- “Benign Overfitting from KKT Conditions for Margin Maximization”
— Conference on Learning Theory, Bangalore *July 2023*
- “Implicit Regularization and Benign Overfitting for Neural Networks in High Dimensions”
— Youth in High Dimensions Workshop, International Center for Theoretical Physics, Trieste *May 2023*
— UC Berkeley, Department of Biostatistics Seminar *April 2023*
— University of British Columbia, Mathematics of Information, Learning, and Data Seminar *January 2023*
— University of Alberta, Dept. of Mathematical and Statistical Sciences, Statistics Seminar *October 2022*
— EPFL, Fundamentals of Learning and Artificial Intelligence Seminar *September 2022*
- “Statistical and Computational Phenomena in Deep Learning”
— University of Wisconsin–Madison, Department of Statistics Seminar[∇] *February 2023*
— University of British Columbia, Department of Statistics Seminar *February 2023*
— University of Illinois, Urbana-Champaign, Department of Statistics Seminar *February 2023*
— University of Michigan, Department of Statistics Seminar *January 2023*
— University of British Columbia, Department of Mathematics Colloquium *January 2023*
— Northwestern University, Department of Statistics and Data Science Seminar[∇] *January 2023*
— University of California, Davis, Department of Statistics Seminar *January 2023*
— University of California, Davis, Department of Mathematics Colloquium *January 2023*
— Boston University, Department of Mathematics and Statistics Seminar *January 2023*
— University of California, San Diego, Department of Mathematics Colloquium *December 2022*
— University of California, Irvine, Department of Statistics Seminar *December 2022*

- “Statistical Learning Theory, Optimization, and Neural Networks” (two hour tutorial)
— Simons Institute, Deep Learning Theory Workshop & Summer School August 2022
- “Benign Overfitting without Linearity”
— University of Southern California, Symposium on Frontiers of Machine Learning and AI November 2022
— Joint Statistical Meetings, Washington, D.C. August 2022
— Conference on Learning Theory, London July 2022
— ETH Zürich, Data, Algorithms, Combinatorics, & Optimization Seminar June 2022
— Harvard University, Probabilistic Seminar[✓] May 2022
— University of Toronto, Statistics Research Day[✓] May 2022
— University of British Columbia, Christos Thrampoulidis group meeting[✓] April 2022
— Theory of Overparameterized Machine Learning Workshop[✓] April 2022
— Google Research, Algorithms Seminar[✓] March 2022
— Oxford University, Yee Whye Teh group meeting[✓] March 2022
— NSF/Simons Mathematics of Deep Learning Seminar[✓] March 2022
- “Random Feature Amplification: Feature Learning and Generalization in Neural Networks”
— EPFL, Nicolas Flammarion group meeting October 2022
— Microsoft Research, Machine Learning Foundations Seminar[✓] April 2022
— Columbia University, Daniel Hsu group meeting[✓] April 2022
— Theory of Overparameterized Machine Learning Workshop[✓] April 2022
- “Self-Training Converts Weak Learners to Strong Learners in Mixture Models”
— Simons Institute for the Theory of Computing, Deep Learning Theory Symposium December 2021
- “Proxy Convexity: A Unified Optimization Framework For Neural Networks Trained by Gradient Descent”
— Simons Institute for the Theory of Computing, Meet the Fellows Welcome Event September 2021
- “Generalization of SGD-Trained Neural Networks in the Presence of Adversarial Label Noise”
— ETH Zürich, Young Data Science Researchers Seminar[✓] April 2021
— Johns Hopkins University, Machine Learning Seminar[✓] April 2021
— Max-Planck-Institute MiS, Machine Learning Seminar[✓] March 2021
— NSF/Simons Mathematics of Deep Learning Seminar[✓] February 2021

PROFESSIONAL SERVICE

- Workshop co-organizer: Deep Learning Theory Workshop & Summer School, Simons Institute for the Theory of Computing, Berkeley, 2022.
- Area Chair/Senior PC for conferences: NeurIPS 2023, ALT 2024, NeurIPS 2024, ALT 2025, ICML 2025, ICLR 2026.
- Area Chair for workshops: Theoretical Foundations of Foundation Models @ ICML 2024.
- Reviewer for journals: *Annals of Statistics*, *Journal of Machine Learning Research*, *Bernoulli*, *SIAM Journal on Mathematics of Data Science*, *Neural Computation*, *Mathematics of Operations Research*, *Transactions on Machine Learning Research*.
- Reviewer for conferences: ICML 2020, NeurIPS 2020, AISTATS 2021, ICML 2021, NeurIPS 2021, ICLR 2022, AISTATS 2022, ICML 2022, ICLR 2023, COLT 2023, ICLR 2025.
- Reviewer for workshops: Theory of Overparameterized Machine Learning (TOPML) 2021, ICML Workshop on Overparameterization: Pitfalls & Opportunities (ICMLOPPO) 2021, TOPML 2022, NeurIPS Workshop on Mathematics of Modern Machine Learning (M3L) 2023, NeurIPS Workshop on Robustness of Few-Shot Learning in Foundation Models (R0-FoMo) 2023.
- Reviewer, ENVISION Research Competition for Women in STEM, 2022.
- Volunteer for Queer in AI.

CONFERENCE, WORKSHOP, AND PROGRAM PARTICIPATION

- MFO Workshop on Modern and Emerging Phenomena in Machine Learning. Oberwolfach, 2026.
- Modern Paradigms in Generalization Program (Long-term Participant), Simons Institute, Berkeley, 2024.
- International Conference on Learning Representations. Vienna, 2024.
- Neural Information Processing Systems. New Orleans, 2023.
- Conference on Learning Theory. Bangalore, 2023.
- Youth in High Dimensions. Trieste, 2023.
- Joint Statistical Meetings. Washington, DC, 2022.
- Deep Learning Theory Summer Cluster. Simons Institute for the Theory of Computing, Berkeley, 2022.
- Conference on Learning Theory. London, 2022.
- Deep Learning Theory Symposium. Simons Institute for the Theory of Computing, Berkeley, 2021.
- Mathematical/Scientific Foundations of Deep Learning Annual Meeting. Simons Foundation, New York, 2021.
- ICML Workshop on Overparameterization: Pitfalls and Opportunities. Online, 2021.
- Theory of Overparameterized Machine Learning (TOPML) Workshop. Rice University (online), 2021; 2022.
- Theory of Deep Learning Special Quarter. TTIC/Northwestern Institute for Data, Econometrics, Algorithms, and Learning (online) 2020.
- Neural Information Processing Systems. Vancouver, BC, 2019.
- Emerging Challenges in Deep Learning Workshop. Simons Institute, Berkeley, 2019.
- Summer School in Probability. Pacific Institute for the Mathematical Sciences, Vancouver, BC, 2014.

TEACHING EXPERIENCE

- UC Davis, Department of Statistics
 - Instructor, Stats 35B: Statistical Data Science II, Winter 2024.
 - Instructor, Stats 250: Theoretical Foundations of Modern AI, Winter 2024.
 - Organizer, Stats 290: Seminar in Statistics. Winter 2024.
- UCLA, Department of Statistics
 - TA, Stats 100C: Linear Models, Spring 2020.
 - TA, Stats 102C: Monte Carlo Methods, Fall 2019.
 - TA, Stats 100B: Mathematical Statistics, Winter 2016.
 - TA, Stats 100A: Probability Theory, Fall 2016.
 - TA, Stats 10: Intro to Statistics, Summer 2016.

PROGRAMMING LANGUAGES AND SKILLS

Python, JAX with TPUs, PyTorch with GPUs, R, Matlab.